

8. FOURIER TRANSFORM AND FREQUENCY DOMAIN ANALYSIS

Objective

This experiment is divided into four simulations:

1. Fourier Transform of Periodic Impulse Train
2. Discrete Time Fourier Transform
3. Discrete Time Fourier Transform of Rectangular Pulse
4. Realization of DIT FFT for Spectral Analysis

Experiment 8.1

Fourier Transform of Periodic Impulse Train

Program

% Test Case: Fourier Transform of a Periodic Impulse Train

```
clc;
```

```
clear;
```

```
close all;
```

```
% Define time range and sampling interval
```

```
t_min = -5;
```

```
t_max = 5;
```

```
dt = 0.01;
```

```
t = t_min:dt:t_max;
```

```
% Define periodic impulse train
```

```
T = 1;      % Period of impulse train
```

```
A = 1;      % Amplitude
```

```
% Create impulse train using tolerance
```

```
x_t = double(abs(mod(t,T)) < 1e-10);
```

```
% Compute Fourier Transform using FFT
```

```
X_f = fft(x_t);
```

```
% Number of samples
```

```
N = length(t);
```

```
% Correct frequency vector
```

```
f = (-N/2:N/2-1)/(N*dt);
```

```
% Shift zero-frequency component to center
```

```
X_f_shifted = fftshift(X_f);
```

```
% Magnitude of Fourier Transform
```

```
magnitude_X_f = abs(X_f_shifted)/N;
```

```
% Phase of Fourier Transform
```

```
phase_X_f = angle(X_f_shifted);
```

```
% Plot the original periodic impulse train
```

```
figure;
```

```
subplot(3,1,1);
```

```
stem(t, x_t, 'b', 'filled');
```

```
xlabel('t (Time)');
```

```
ylabel('Amplitude');
```

```
title('Periodic Impulse Train');
```

```
grid on;
```

```
% Plot magnitude spectrum
```

```
subplot(3,1,2);
```

```
plot(f, magnitude_X_f, 'r');
```

```
xlabel('f (Frequency)');
```

```
ylabel('|X(f)|');
```

```
title('Magnitude of Fourier Transform');
```

```
grid on;
```

```
% Plot phase spectrum
```

```
subplot(3,1,3);
```

```
plot(f, phase_X_f, 'g');
```

```
xlabel('f (Frequency)');
```

```
ylabel('Phase of X(f)');
```

```
title('Phase of Fourier Transform');
```

```
grid on;
```

OUTPUT



Experiment 8.2

Discrete Time Fourier Transform

Program

% Test Case: DTFT of a Discrete-Time Signal

```
clc;
```

```
clear;
```

```
close all;
```

```
% Define parameters
```

```
N = 50;          % Number of samples
```

```
f0 = 0.1;        % Frequency component 1
```

```
f1 = 0.3;        % Frequency component 2
```

```
n = 0:N-1;      % Sample index
```

```
% Define the signal
```

```
x = cos(2*pi*f0*n) + sin(2*pi*f1*n);
```

```
% Compute DTFT
```

```
omega = linspace(-pi, pi, 1000); % Frequency range
```

```
% DTFT calculation
```

```
X = zeros(size(omega));
```

```
for k = 1:length(omega)
```

```
    X(k) = sum(x .* exp(-1j * omega(k) * n));
```

```
end
```

```
% Plot the magnitude of DTFT
```

```
figure;
```

```
plot(omega/pi, abs(X), 'LineWidth', 1.5);
```

```
xlabel('Normalized Frequency (\omega/\pi)');
```

```
ylabel('|X(e^{j\omega})|');
```

```
title('Magnitude of DTFT');
```

```
grid on;
```

```
% Plot the phase of DTFT
```

```
figure;
```

```
plot(omega/pi, angle(X), 'LineWidth', 1.5);
```

```
xlabel('Normalized Frequency (\omega/\pi)');
```

```
ylabel('Phase of X(e^{j\omega})');
```

```
title('Phase of DTFT');
```

```
grid on;
```

OUTPUT

